

A Design Methodology for Disturbance Observer with Application to Precision Motion Control: An H-infinity Based Approach

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Abstract—This paper proposes a disturbance observer (DOB) design procedure for general multi-input-multi-output (MIMO) systems based on the H-infinity synthesis. The DOB is designed by minimizing the H-infinity norm of the dynamic system from the disturbance to its estimation error. The design procedure does not require an explicit plant inverse. The proposed DOB is applied to a dual-stage hard disk drive (HDD) whose plant inverse is not applicable because of its dual-input-single-output nature. Simulation results show that the disturbance is effectively estimated and compensated enhancing precision of hard disk drive (HDD) servo for positioning of the recording head.

I. INTRODUCTION

Disturbance observer (DOB) is a powerful technique to estimate and compensate the disturbance in high-precision systems. It is a plant-inverse based technique and has many industrial applications such as hard disk drives (HDDs) [1], [2], power-assist electric bicycles [3], wafer scanning systems [4], manipulators [5], and autonomous vehicles [6]. DOB is usually combined with other control techniques such as adaptive control [7], repetitive control [8], and iterative learning control [4] to improve system performances. The general DOB design procedure includes two steps: (1) design a stable inverse of the plant; and (2) design a Q-filter to maintain the causality and robustness. The plant inverse is usually obtained from a low-order nominal model of the plant. The Q-filter can be designed as a low/high/band-pass filter based on the frequency characteristics of the disturbance and the uncertainties in the plant.

Robust control theory has been incorporated into various algorithms to guarantee the stability and the robustness of control in high precision systems. These algorithms include anti-windup scheme [9], sliding mode control [10], iterative learning control [11], [12], and DOB [13]–[16]. In [13], the DOB design problem was transformed into the H-infinity synthesis problem by finding an optimal static output feedback gain for an extended plant. In [14]–[16], the Q-filter was designed through solving an H-infinity optimization problem. However, these DOB design procedures were only applied to SISO systems and were based on the conventional DOB structure that requires a well-designed stable inverse of plant. Usually designing a stable plant inverse is not trivial for some SISO systems and even more challenging for MIMO systems.

There exist some DOB design methods for MIMO systems in the literature. For example, the method in [17] treated each input-output channel of the plant separately by ignoring the coupling effect. However, this introduced plant modeling errors and the stability was difficult to guarantee. An alternative method proposed in [18] first decoupled the system using the nominal model of the plant and then followed the conventional DOB design procedure for SISO systems. These techniques did not mitigate the issue of designing a good plant inverse in the DOB design. Furthermore, most of them were only applicable to the square systems: systems with the same dimensions of the inputs and outputs. It is rather challenging to apply these DOB techniques to the systems with the inputs of higher dimension than the outputs.

To unnesessitate plant inverse and apply DOB to a general class of MIMO plant, instead of following the conventional DOB structure, this paper formulates the DOB design problem into an H-infinity optimization problem by treating the whole observer as a ‘black’ box without specifying any explicit structure. The proposed design methodology is different from the existing H-infinity design methods for DOBs which still follow the conventional DOB structure with well-designed plant inverse and Q-filters. Being relaxed from the restrictions due to the conventional the DOB structure, the proposed approach has more design flexibilities and is possible to achieve better performance than the conventional DOB. It is also worth mentioning that the proposed DOB is still an add-on algorithm aiming to estimate and compensate the disturbance without redesigning the baseline feedback controller.

The remainder of the paper is organized as follows: Section II introduces the conventional DOB and its design methodology; Section III formulates the DOB design problem into an H-infinity optimization problem; Section IV applies the proposed DOB onto dual-stage HDDs; Section V provides the validation of the proposed method by simulation and Section VI concludes the paper.

II. CONVENTIONAL DOB DESIGN METHODOLOGY

Figure 1 shows a general system with a conventional DOB (in the dash rectangular box), in which $G(z)$ is the plant; $C(z)$ is the baseline feedback controller; $G_n(z)$ is a nominal model of the plant; and $Q(z)$ is a filter to maintain the causality and robustness of the DOB. The reference signal r , the output signal y , the control signal u generated by $C(z)$, the disturbance d , the disturbance estimate $\hat{d}$, and the control signal $u_e = u - \hat{d}$ injected into the plant are all defined in this

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figure. The DOB includes a plant inverse $G_n^{-1}(z)$ and a filter $Q(z)$; the input signals to the DOB are u_e and y , and the output signal is $\hat{d}$.

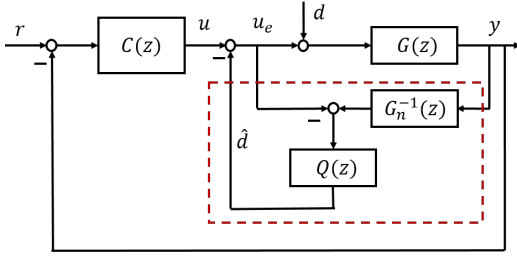


Fig. 1. A general system with conventional DOB

The intuitive idea of the conventional DOB is to utilize the inverse of the plant model to reconstruct the plant's input signals which consist of the control signal u_e and the actual disturbance d . With well-defined $G_n^{-1}(z)$ and $Q(z)$, the DOB can recover the actual disturbance over specified frequency ranges. To explain this, the transfer function from d to $\hat{d}$ is derived as follows. From Figure 1,

$$\hat{d} = Q(z)[G_n^{-1}(z)y - (u - \hat{d})] \quad (1)$$

which implies

$$(1 - Q(z))\hat{d} = Q(z)G_n^{-1}(z)y - Q(z)u \quad (2)$$

Noting that $y = G(z)[u - \hat{d} + d]$ and $u = -C(z)y$,

$$y = G(z)[-C(z)y - \hat{d} + d] \quad (3)$$

which implies

$$y = [1 + G(z)C(z)]^{-1}G(z)(d - \hat{d}) \quad (4)$$

Substituting $u = -C(z)y$ and Eq. (4) into Eq. (1), after some manipulations, we have

$$\hat{d} = \frac{Q[G_n^{-1}G + GC][1 + GC]^{-1}}{(1 - Q) + Q[G_n^{-1}G + GC][1 + GC]^{-1}} d \quad (5)$$

where z is omitted for simplicity. Note that if

$$Q(z) = 1 \quad (6)$$

and

$$G_n^{-1}(z)G(z) = 1 \quad (7)$$

then $d = \hat{d}$, which means that the actual disturbance d is completely reconstructed.

In general there are three considerations in DOB design: stability, causality, and robustness. (1) Stability: if $G(z)$ is a non-minimum phase model (i.e., the zeros of $G(z)$ are unstable), there does not exist a stable plant inverse satisfying Eq. (7); an alternative stable plant inverse satisfying $G_n^{-1}(z)G(z) \approx 1$ should be designed. (2) Causality: if $G(z)$ is strictly causal, then its inverse is non-causal and not realizable; a $Q(z)$ needs to be designed such that $Q(z)G_n^{-1}(z)$ is causal. (3) Robustness: if there exist some uncertainties in $G(z)$, i.e., $G(z) = G_n(z)(1 + \Delta(z))$ where $\Delta(z)$ denotes bounded un-modeled dynamics, $Q(z)$ needs to be designed such that robustness of the system is guaranteed. Detailed discussions can be found in [19].

III. DOB DESIGN - REFORMULATION

This section formulates the DOB design problem into an H-infinity optimization problem based on robust control theory. Define $T_f(z)$ as the transfer function from d to $(d - \hat{d})$, i.e.,

$$d - \hat{d} = T_f(z)d \quad (8)$$

From Eq. (5),

$$T_f = 1 - \frac{Q[G_n^{-1}G + GC][1 + GC]^{-1}}{(1 - Q) + Q[G_n^{-1}G + GC][1 + GC]^{-1}} \quad (9)$$

In this section, the DOB design problem is transferred into the minimization of the H-infinity norm of $T_f(z)$, i.e., $\|T_f(z)\|_\infty$, which is defined as the supremum of the maximum singular value of $T_f(e^{j\Theta})$ ($\Theta \in [0, 2\pi)$),

$$\|T_f(z)\|_\infty = \sup_{\Theta \in [0, 2\pi)} \bar{\sigma}[T_f(e^{j\Theta})] \quad (10)$$

where $\bar{\sigma}[\cdot]$ denotes the maximum singular value of a matrix. If $\bar{\sigma}[T_f(e^{j\Omega})]$ is small over certain frequency range, $(d - \hat{d})$ would be small, and thus $\hat{d}$ is a good estimate of d over this frequency range.

Assume $T_f(z)$ has the following state-space realization:

$$T_f : \begin{bmatrix} x_c(k+1) \\ d(k) - \hat{d}(k) \end{bmatrix} = \begin{bmatrix} A_c & B_c \\ C_c & D_c \end{bmatrix} \begin{bmatrix} x_c(k) \\ d(k) \end{bmatrix} \quad (11)$$

The following optimization is formulated to achieve the best disturbance estimate in the sense of the smallest $\|T_f(z)\|_\infty$,

$$\begin{aligned} \min_{Q(z), G_n^{-1}(z)} & \|T_f(z)\|_\infty \\ \text{s.t.} & \lambda_i(A_c) \leq 1 \quad \forall i \\ & Q(z)G_n^{-1}(z) \text{ causal} \end{aligned} \quad (12)$$

where $\lambda_i(A_c)$ denotes the i^{th} eigenvalue of A_c . The optimization problem (12) is very difficult to solve: it is not convex with respect to the decision variables $Q(z)$ and $G_n^{-1}(z)$. To transform it into a convex optimization problem, the following new variable $D(z)$ is introduced:

$$D(z) = [D_1(z) \ D_2(z)] = [-Q(z) \ Q(z)G_n^{-1}(z)] \quad (13)$$

This leads to a new expression of $T_f(z)$ with respect to the new variable $D(z)$, as the following proposition describes.

Proposition: With the definition of T_f in Eq. (8), and the definition of D in Eq. (13), T_f can be written as

$$T_f = F_l(M, D) = M_{11} + M_{12}D(I - \begin{bmatrix} M_{22} \\ M_{32} \end{bmatrix} D)^{-1} \begin{bmatrix} M_{21} \\ M_{31} \end{bmatrix} \quad (14)$$

where F_l stands for the linear fractional transformation (LFT), and

$$M = \begin{bmatrix} 1 & -1 \\ -C(1 + GC)^{-1}G & 1 - C(1 + GC)^{-1}G \\ (1 + GC)^{-1}G & (1 + GC)^{-1}G \end{bmatrix} \quad (15)$$

Proof: Figure 2 is an equivalence of the system in Figure 1. M is the transfer function from $[d, \hat{d}]^T$ to $[d - \hat{d}, u_e, y]^T$. Therefore, $T_f = F_l(M, D)$ as described in Eq. (14)

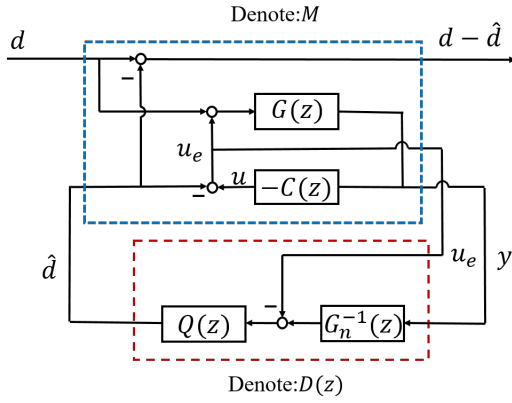


Fig. 2. Equivalent representation for the system in Fig. 1

Based on the Proposition, the optimization problem (12) is reformulated as

$$\begin{aligned} \min_{D(z), \text{causal}} \quad & \|F_l(M(z), D(z))\|_\infty \\ \text{s.t.} \quad & \lambda_i(A_c) \leq 1 \quad \forall i \\ & D_2(z) = -D_1(z)G_n^{-1}(z) \end{aligned} \quad (16)$$

for which the decision variable becomes $D(z)$ with the constraint of $D_2(z) = -D_1(z)G_n^{-1}(z)$. This constraint is a requirement induced from the conventional structure of DOB. In this paper, this constraint is relaxed to utilize H-infinity synthesis and to provide more flexibilities in the DOB design. With the constraint relaxing, the optimization problem becomes

$$\begin{aligned} \min_{D(z) \text{ causal}, \gamma} \quad & \gamma \\ \text{s.t.} \quad & |\lambda_i(A_c)| < 1 \quad \forall i \\ & \|F_l(M(z), D(z))\|_\infty < \gamma \end{aligned} \quad (17)$$

Compared to the conventional DOB design, the proposed DOB design solves the optimization problem (17) over a larger feasible region with the constraint relaxing, which results in a smaller γ . This optimization problem can be reformulated into a convex optimization problem with linear matrix inequality (LMI) constraints [20] which can be solved efficiently thereafter.

Remarks: The main differences between Problem (12) and Problem (17) arise from two aspects: (a) Problem (17) considers D as the decision variable, while Problem (12) considers Q and G_n^{-1} as the decision variables; and (b) Problem (17) removes the constraint for D . With variable transforming and constraint relaxing, Problem (17) can be reformulated into a convex optimization problem with LMI constraints based on robust control theory. Furthermore, the proposed DOB design procedure can be modified into the one that explicitly considers system uncertainties based on μ synthesis.

IV. APPLICATION TO DUAL-STAGE HDDS

In the literature, the DOB design for dual-stage HDDs has not been fully investigated due to its dual-input-single-output plant model whose inverse does not exist. Using the proposed DOB design procedure, the disturbance can be estimated and

compensated in both the voice coil motor (VCM) loop and the piezoelectric motor (PZT) loop.

The classic dual-stage HDD control scheme is shown in Figure 3 [21], in which P_v is the VCM plant, P_m is the PZT plant, C_v is the baseline feedback controller for P_v , C_m is the baseline feedback controller for P_m , and $\hat{P}_m$ is the nominal plant of P_m . The signals are also defined in Figure 3: r is the reference, y is the output, d_v is the disturbance in the VCM loop, d_m is the disturbance in the PZT loop, and u_v and u_m are the control signals generated by C_v and C_m respectively. The position error signal (PES) is defined as $e = r - y$.

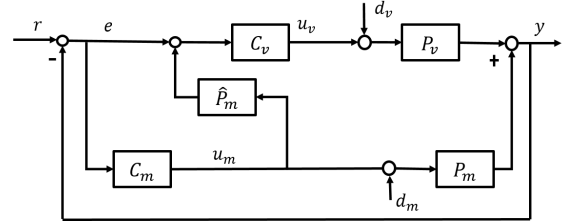


Fig. 3. Dual-stage HDD

The H-infinity based DOB (denoted as D) is designed as shown in Figure 4. The input signals to D are u_{ve} , u_{me} , and y ; the output signals from D are $\hat{d}_v$ and $\hat{d}_m$; $u_{ve} = u_v - \hat{d}_v$ and $u_{me} = u_m - \hat{d}_m$.

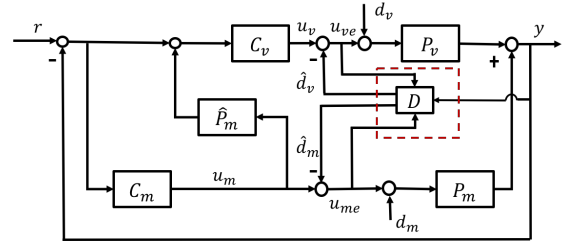


Fig. 4. Dual-stage HDD with DOB

The scheme in Figure 4 is further transformed to Figure 5 for simplicity. Separating the design parameter D from other dynamics in Figure 5, an H-infinity based DOB scheme can be constructed in Figure 6, where C and P are

$$C = \begin{bmatrix} (1 + C_m \hat{P}_m) C_v & 0 \\ 0 & C_m \end{bmatrix}, \quad P = \begin{bmatrix} P_v & 0 \\ 0 & P_m \end{bmatrix} \quad (18)$$

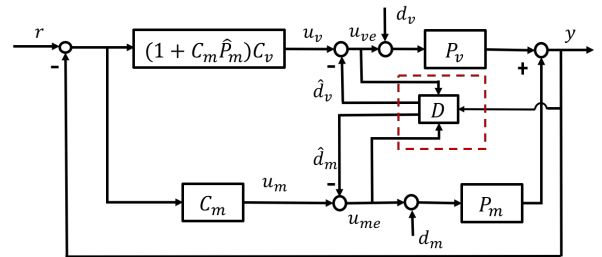


Fig. 5. Equivalent structure of Fig. 4

Denote

$$d = \begin{bmatrix} d_m \\ d_v \end{bmatrix}, \quad \hat{d} = \begin{bmatrix} \hat{d}_m \\ \hat{d}_v \end{bmatrix}, \quad u_e = \begin{bmatrix} u_{ve} \\ u_{me} \end{bmatrix} \quad (19)$$

B. DISO case

In the second simulation study, the proposed DOB is applied to a dual-stage HDD benchmark model [23]. The weighting filters $W_v(z)$ and $W_m(z)$ are designed based on the frequency characteristics of the disturbance in both the VCM and the PZT loops. Similar to the assumption made in [18], it is assumed that the disturbance in the VCM loop focuses around 1000 Hz and that the disturbance in the PZT loop focuses around 2500 Hz. Therefore, $W_v(z)$ and $W_m(z)$ are designed as band-pass filters or peak filters centered around 1000 Hz and 2500 Hz respectively. Figure 12 provides the bode plots of $W_v(z)$ and $W_m(z)$ which are used in this simulation study.

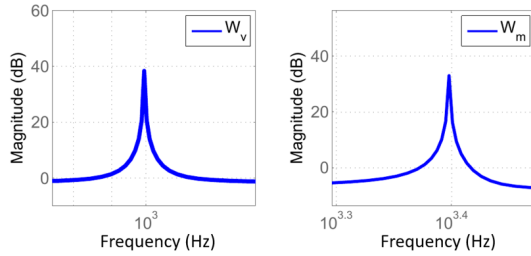


Fig. 12. Bode plots of weighting filters

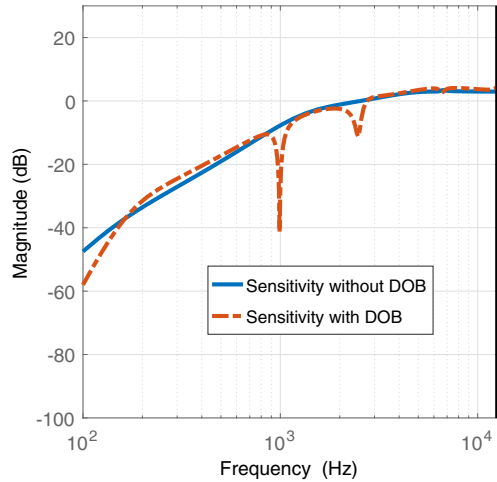


Fig. 13. Bode plots of sensitivities with and without DOB

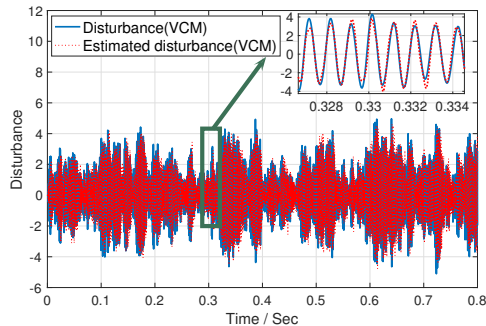


Fig. 14. Disturbance estimate: d_v and $\hat{d}_v$

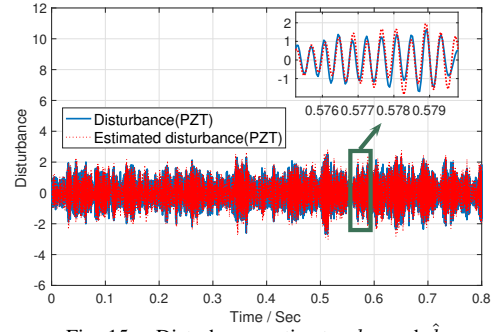


Fig. 15. Disturbance estimate: d_m and $\hat{d}_m$

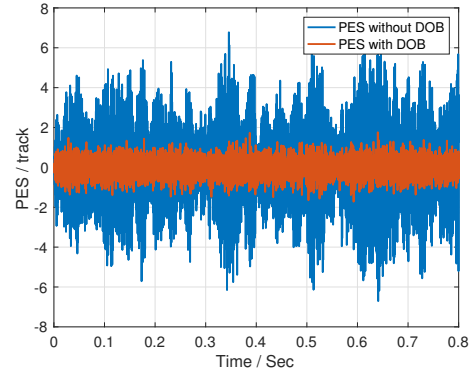


Fig. 16. PES comparison with and without DOB (time domain)

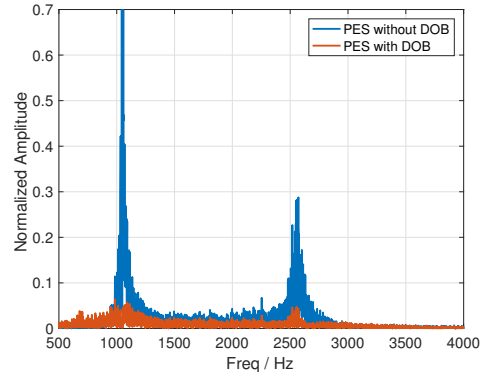


Fig. 17. PES comparison with and without DOB (frequency domain)

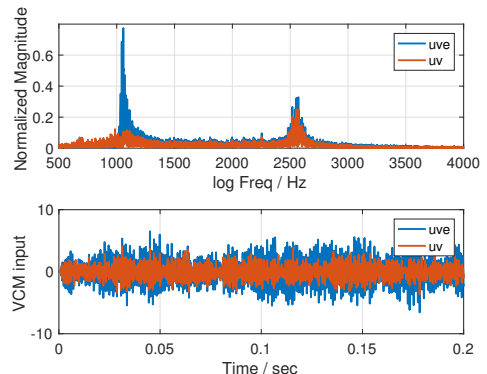


Fig. 18. Control signal in VCM loop

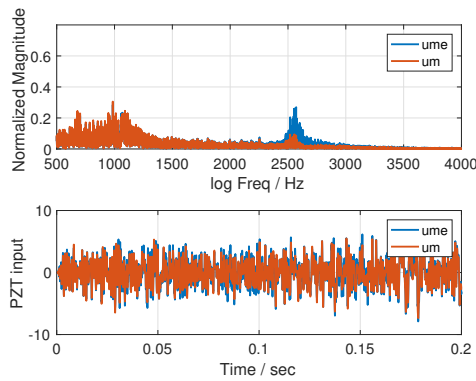


Fig. 19. Control signal in PZT loop

Simulation results are provided in Figures 13-19. Figure 13 compares the bode plots of the system sensitivities with and without the proposed DOB. It is observed that the dual-stage HDD with the proposed DOB has better vibration attenuation around 1000 Hz and 2500 Hz. Figures 14 to 17 show a practical case when the input disturbance data is modified from actual on-drive test. Figures 14 and 15 provide the disturbance estimates: the proposed DOB is able to estimate the disturbance in both the VCM loop and the PZT loop. The PES comparisons (with and without DOB) in both time domain and frequency domain are provided in Figures 16 and 17 respectively. It is observed from Figure 17 that the amplitude of the PES around 1000 Hz and 2500 Hz has been significantly reduced. Figures 18 and 19 provide the control signals in VCM and PZT loops in both time domain and frequency domain. It is worth noticed that, with the proposed DOB, the control effort around 1000 Hz increases in the VCM loop, while the one around 2500 Hz increases in the PZT loop.

VI. CONCLUSIONS AND FUTURE WORK

A generalized DOB design procedure has been proposed for both SISO systems and MIMO systems based on H-infinity synthesis. The proposed DOB assures the stability and minimizes the weighted H-infinity norm of the dynamics from the disturbance to its estimation error. This DOB design procedure is applicable not only to the square systems, but also to the systems with the inputs of higher dimension than the outputs. Detailed evaluation has been performed on a dual-stage HDD plant that has dual inputs and single output. The simulation results demonstrate the effectiveness of the proposed DOB. As a future exploration, this design procedure will be modified to explicitly take the system variations into consideration.

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